

SUBSTATION CONDITION

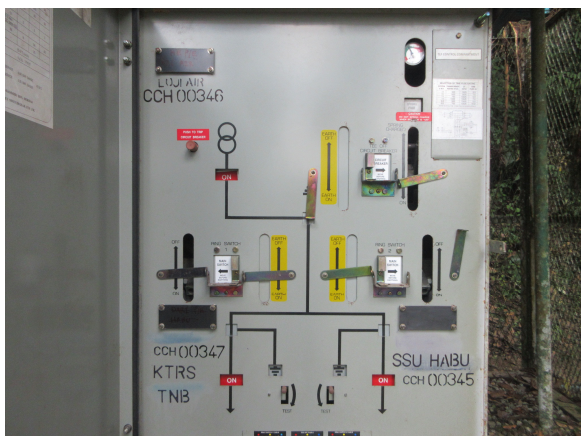
SUBSTATION OVERVIEW



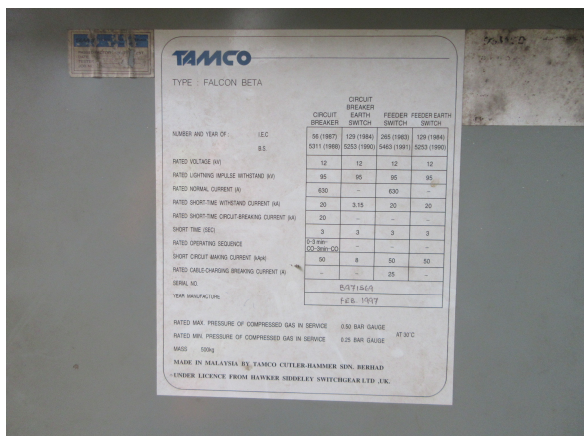
SIGNBOARD



SWITCHGEAR



SWITCHGEAR NAMEPLATE



EFI



SF6 INDICATOR



VISUAL DEFECT

SUBSTATION



CPR POSTER OLD VERSION – SWG BAY

NORMAL/DEFECT STICKER

NORMAL STICKER	
CONDITION ASSESSMENT  TENAGA NASIONAL <i>Enter Right</i> DIAGNOSTIC UNIT SUBSTATION 	
Cycle/Year :	1/2026
Testing Date :	01-06-2026
Defect :	—
Company/Team :	EET ENGINEERING
	TEV U/S IR

152
SS

STATE / AREA: PAHANG / CAMERON HIGHLAND										DISTRIBUTION SUBSTATION (PCS) CMA TEST SHEET										DATE: 01/06/2026										FUNCTIONAL LOCATION: CCHL/PCB/1700155																			
PE NAME: SIS LOJI AIR HABU										TIME IN / OUT: 4:21 pm - 4:34 pm										TEV BACK GROUND / HUMIDITY: 3 - 80 %										BACKGROUND TEMP: 21.2 °C																			
V.I REF. NO:										SWITCHGEAR										TEMPERATURE										ULTRASOUND										TEV									
NO.	FDR. NO	FEEDER NAME	STATUS (TRIP / CLOSE)	LOAD (A)	CABLE TYPE/SIZE	HEATER (A)	SERIAL NO (ONLY BREAKER & OLIV)	COMP	Tmin (°C)	Tmax (°C)	AT	Avg (°C)	FILE	REMARKS	dB	FILE	TYPE CHAR.	dB	PULSE / CYCLE	TYPE CHAR.	REMARK																												
1	CCH 00347	KTR5 TNB	TRIP / CLOSE		XLPE PILC N/A NACC			CABLE	27.0	27.2	0.2	27.1	39.30		0	7	-	4	0	-																													
2	CCH 00346	LOJI AIR	TRIP / CLOSE		XLPE PILC N/A NACC			CABLE	26.5	26.9	0.4	26.8	31.32		0	8	-	4	0	-																													
3	CCH 00345	SSV HABU	TRIP / CLOSE		XLPE PILC N/A NACC			CABLE	26.4	26.7	0.3	26.5	33.34		0	9	-	5	0	-																													
4			TRIP / CLOSE		XLPE PILC N/A NACC			CABLE																																									
OVERVIEW										bottom										u6																													

LOCAL TRANSFORMER										THERMOGRAPHY										ULTRASOUND										REMARKS									
TX ID	SERIAL NO	CABLE TYPE/SIZE	REMARK	COMP.	Tmin(°C)	Tmax(°C)	AT	Avg (°C)	FILE	dB	TYPE CHAR.	TX ID	SERIAL NO	CABLE TYPE/SIZE	REMARK	COMP.	Tmin(°C)	Tmax(°C)	AT	Avg (°C)	FILE	dB	TYPE CHAR.	TX ID	SERIAL NO	CABLE TYPE/SIZE	REMARK	COMP.	Tmin(°C)	Tmax(°C)	AT	Avg (°C)	FILE	dB	TYPE CHAR.	REMARKS			
TX1	N/A	XLPE/PILC		HT CABLE								TX3		XLPE/PILC		HT CABLE											XLPE/PILC		HT CABLE										
		XLPE/PILC		HT BUSHING										XLPE/PILC		HT BUSHING											XLPE/PILC		HT BUSHING										
		XLPE/PILC		LV CABLE										XLPE/PILC		LV CABLE											XLPE/PILC		LV CABLE										
		XLPE/PILC		LV BUSHING										XLPE/PILC		LV BUSHING											XLPE/PILC		LV BUSHING										
		XLPE/PILC		BODY										XLPE/PILC		BODY											XLPE/PILC		BODY										
		XLPE/PILC		HT CABLE										XLPE/PILC		HT CABLE											XLPE/PILC		HT CABLE										
		XLPE/PILC		HT BUSHING										XLPE/PILC		HT BUSHING											XLPE/PILC		HT BUSHING										
		XLPE/PILC		LV CABLE										XLPE/PILC		LV CABLE											XLPE/PILC		LV CABLE										
		XLPE/PILC		LV BUSHING										XLPE/PILC		LV BUSHING											XLPE/PILC		LV BUSHING										
		XLPE/PILC		BODY										XLPE/PILC		BODY											XLPE/PILC		BODY										

LV DISTRIBUTION BOARD / FEEDER PILLAR / OTHERS										THERMOGRAPHY										ULTRASOUND										REMARKS									
NO	LVDB / FP	DEFECTIVE FEEDER NAME	Tmin(°C)	Tmax(°C)	AT	FILE	Avg (°C)	REMARK	LVDB / FP 1	LVDB / FP 2	LVDB / FP 3	LVDB / FP 4	LVDB / FP 5	LVDB / FP 6	LVDB / FP 7	LVDB / FP 8	LVDB / FP 9	LVDB / FP 10	LVDB / FP 11	LVDB / FP 12	LVDB / FP 13	LVDB / FP 14	LVDB / FP 15	LVDB / FP 16	LVDB / FP 17	LVDB / FP 18	LVDB / FP 19	LVDB / FP 20	LVDB / FP 21	LVDB / FP 22	LVDB / FP 23	LVDB / FP 24	LVDB / FP 25	LVDB / FP 26	LVDB / FP 27	LVDB / FP 28	LVDB / FP 29	LVDB / FP 30	
1	LVDB / FP 1	INC. 1							F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23	F24	F25	F26	F27	F28	F29	F30	
2	LVDB / FP 2	INC. 2							XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC	XLPE/PILC		

NO	LVDB / FP	DEFECTIVE FEEDER NAME	Tmin(°C)	Tmax(°C)	AT	FILE	Avg (°C)	REMARK	LVDB / FP 1	LVDB / FP 2	LVDB / FP 3	LVDB / FP 4	LVDB / FP 5	LVDB / FP 6	LVDB / FP 7	LVDB / FP 8	LVDB / FP 9	LVDB / FP 10	LVDB / FP 11	LVDB / FP 12	LVDB / FP 13	LVDB / FP 14	LVDB / FP 15	LVDB / FP 16	LVDB / FP 17	LVDB / FP 18	LVDB / FP 19	LVDB / FP 20	LVDB / FP 21	LVDB / FP 22	LVDB / FP 23	LVDB / FP 24	LVDB / FP 25	LVDB / FP 26	LVDB / FP 27	LVDB / FP 28	LVDB / FP 29	LVDB / FP 30
1	INC 1																																					
2	INC 2																																					
3	F1																																					
4	F2																																					
5	F3																																					
6	F4																																					
7	F5																																					
8	F6																																					
9	F7																																					
10	F8																																					

BATTERY BANK										THERMOGRAPHY										ULTRASOUND										REMARKS									
NO	LVDB / FP	DEFECTIVE FEEDER NAME	Tmin(°C)	Tmax(°C)	AT	FILE	Avg (°C)	REMARK	LVDB / FP 1	LVDB / FP 2	LVDB / FP 3	LVDB / FP 4	LVDB / FP 5	LVDB / FP 6	LVDB / FP 7	LVDB / FP 8	LVDB / FP 9	LVDB / FP 10	LVDB / FP 11	LVDB / FP 12	LVDB / FP 13	LVDB / FP 14	LVDB / FP 15	LVDB / FP 16	LVDB / FP 17	LVDB / FP 18	LVDB / FP 19	LVDB / FP 20	LVDB / FP 21	LVDB / FP 22	LVDB / FP 23	LVDB / FP 24	LVDB / FP 25	LVDB / FP 26	LVDB / FP 27	LVDB / FP 28	LVDB / FP 29	LVDB / FP 30	
1	Black Box 1																																						
2	Black Box 2																																						
3	Lightning Arrestor																																						
4	Drypoint Fuse																																						
5	Auto Recloser																																						
6	LBS																																						

EET ENGINEERING SERVICES SDN BHD
 No. 38-2, 2nd Floor, Jalan SS 19/1D
 47500 Subang Jaya, Selangor.
 Tel: 603-5636 6205 Fax: 603-5636 7205

MOHD AZZAD BIN OTHMAN (10085230)
 JURUTEKNIK TINGKATAN BIASA
 (DIAGNOSTIC SUBSTATION PAHANG)
 SBU MAINTENANCE EAST
 DISTRIBUTION NETWORK DIVISION, TNB

152 SS

LPA COV. Sny Dug 1



VISUAL INSPECTION CHECKLIST FOR 11kV SUBSTATION

DATE: 1/6/2026
P/E Name (ERMS): S/S LOJI AIR HABU
P/E Name (SITE): SS LOJI AIR HABU
GPS Coordinate: 4.438017, 101.390712
PCE Type: INDOOR
PCE Area: NORMAL
Switchgear Type 1: RMU OIL
Manufacturer/Modal: TAMCO / FALCON BETA
Switchgear Manufactured Date: 1997
Switchgear Type 2: RMU OIL
Manufacturer/Modal: N/A
Switchgear Manufactured Date: N/A
No. of Transformer: N/A
Tx 1: Type
Tx 2: Type
Tx 3: Type
Tx 4: Type

F/L NO (ERMS): CCLAL/PCE/300155
F/L NO (SITE): CCLAL/PCE/300155

OUTDOOR
INDUSTRIAL
RMU SF6
COMPACT
COASTAL
MRMU
ATTACH BUILDING
PAT
VCB
OCB

Switchgear Rating: 12kV
RMU SF6 / OIL S/N: B971569

Switchgear Rating:
RMU SF6 / OIL S/N:
Construction Yr:
Manufacturer:
S/N:
Construction Yr:
Manufacturer:
S/N:
Construction Yr:
Manufacturer:
S/N:

KEJANGGALAN

1. Signboard:
2. Earth connection:
3. EFI Functioning:
4. Heater Functioning:
5. Labelling @Switchgear:
6. Labelling @ LVDB:
7. Gas Pressure:
8. Tx Guard:
9. Tx Oil Level (conservator tank):
10. Tx Oil Leak:
11. Tx Bushing:
12. Top Demand Indicator (@LVDB):
13. Link/fuse Holder:
14. LV Termination:
15. Rubber Mat:
16. Trenching:
17. Battery Voltage:
18. Lighting:
19. Roof Condition:
20. Fire Fighting
21. Fence/door condition:
22. Cleanliness/Bushes:
23. Vandalisme
24. Other Findings

YES	✓	NO		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
YES	✓	NO		Remarks
Good	✓	Not Good		Remarks
YES	✓	NO		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
YES	✓	NO		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
Valid	✓	Expired		Remarks
Good	✓	Not Good		Remarks
Good	✓	Not Good		Remarks
YES	✓	NO	✓	Remarks

TESTED BY (NAME & SIGNATURE):

EET ENGINEERING SERVICES SDN BHD

No. 38-2, 2nd Floor, Jalan SS 19/1D
47500 Subang Jaya, Selangor.

Tel: 603-5636 6205 Fax: 603-5636 7205

TNB SUPERVISOR/ ENGINEER (NAME & SIGNATURE):

MOHD AZZAD BIN OTHMAN (10085230)
JURUTEKNIK TINGKATAN BIASA
(DIAGNOSTIC SUBSTATION PAHANG)
SBU MAINTENANCE EAST
DISTRIBUTION NETWORK DIVISION, TNB